

Automatic Gain Control (AGC) System using JFET-Based Variable Gain Amplifier

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Objective

To design and simulate an Automatic Gain Control (AGC) system in LTspice capable of maintaining a nearly constant output amplitude over a wide range of input signal amplitudes. The system employs a JFET-based Variable Gain Amplifier (VGA), a precision peak detector, and a feedback control loop for automatic gain regulation.

Theory

Automatic Gain Control (AGC)

Automatic Gain Control is a closed-loop feedback technique that automatically adjusts the gain of an amplifier to maintain a nearly constant output amplitude despite variations in the input signal level. AGC circuits are widely used in communication receivers, audio amplifiers, instrumentation systems, and sensor interfaces.

The AGC loop consists of:

- Variable Gain Amplifier (VGA)
- Precision Peak Detector
- Envelope Detector
- Feedback Control Path

Variable Gain Amplifier (VGA)

The VGA is implemented using an ADTL082 operational amplifier and a 2N3819 N-channel JFET.

The JFET operates as a voltage-controlled resistor in the op-amp feedback path.

Changing the gate voltage changes the drain-source resistance of the JFET, thereby varying the closed-loop gain of the amplifier.

Increasing the negative gate voltage:

- Increases JFET resistance
- Reduces amplifier gain

Decreasing the negative gate voltage:

- Lowers JFET resistance
- Increases amplifier gain

Precision Peak Detector

A precision peak detector built using ADTL082 op-amps, diodes, and an RC network accurately tracks the peak value of the amplifier output while eliminating diode threshold errors.

The capacitor stores the peak voltage, producing the signal envelope.

Feedback Control Loop

The detected envelope is inverted to generate a negative control voltage.

This control voltage is applied to the JFET gate, automatically adjusting the amplifier gain.

The feedback loop continuously regulates the output amplitude, implementing automatic gain control.

Components Used

Software

- LTspice XVII

Operational Amplifiers

- ADTL082 (3 Op-Amps)

Active Components

- 2N3819 N-channel JFET
- Silicon Diodes

Passive Components

- Feedback resistor = 100 k Ω
- Input resistors = 8.2 k Ω
- Envelope capacitor = 1 μ F
- Load resistor = 10 k Ω

Power Supply

- ± 12 V Dual Supply

Input Signal

- Sinusoidal
- Frequency = 1 kHz
- Amplitude varied from 80 mV to 600 mV

Circuit Description

The input signal is first applied to the JFET-based Variable Gain Amplifier.

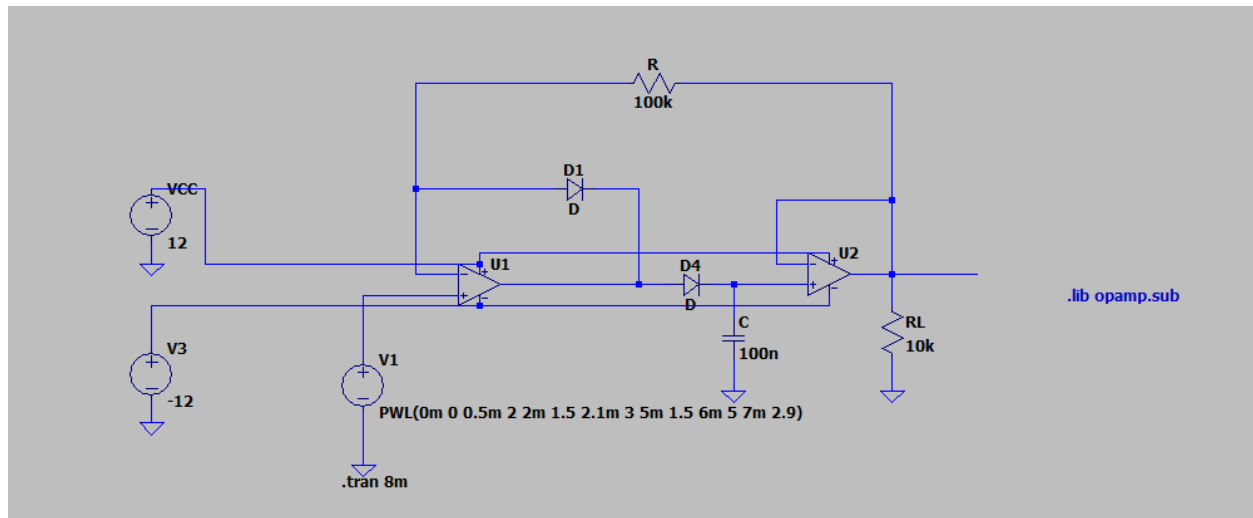
The amplifier output is fed to a precision peak detector, which extracts the signal envelope.

The detected envelope is inverted to generate a negative control voltage.

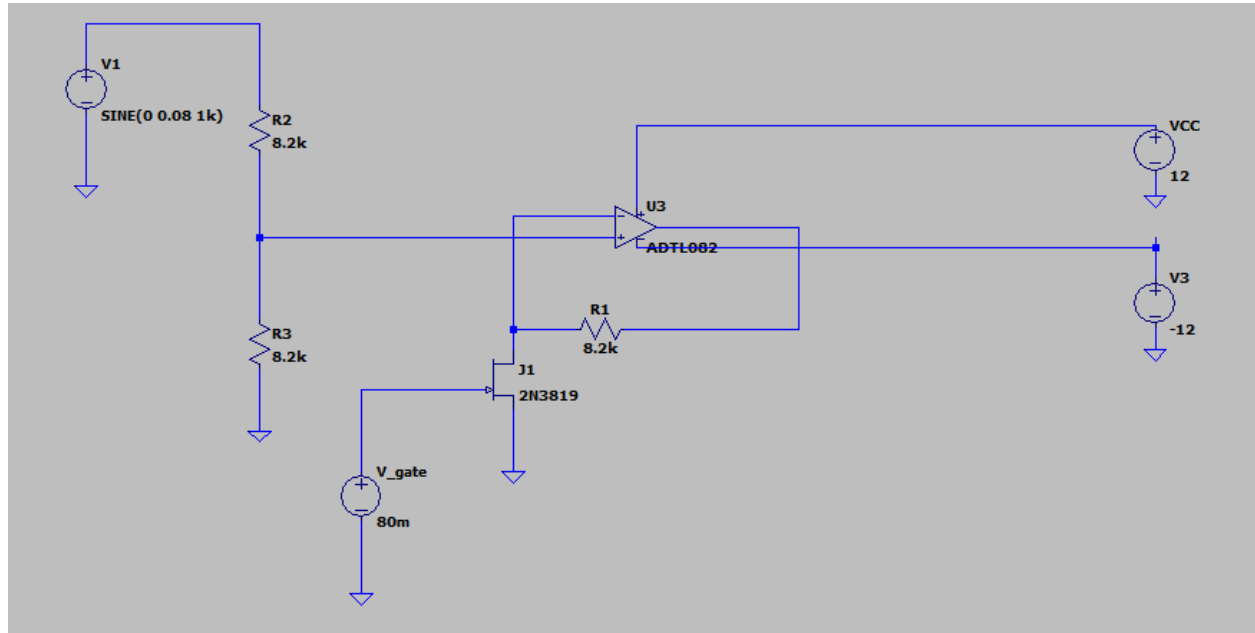
This voltage drives the gate of the 2N3819 JFET, changing its channel resistance and thereby regulating the VGA gain.

The closed-loop system continuously adjusts the gain to maintain an approximately constant output amplitude.

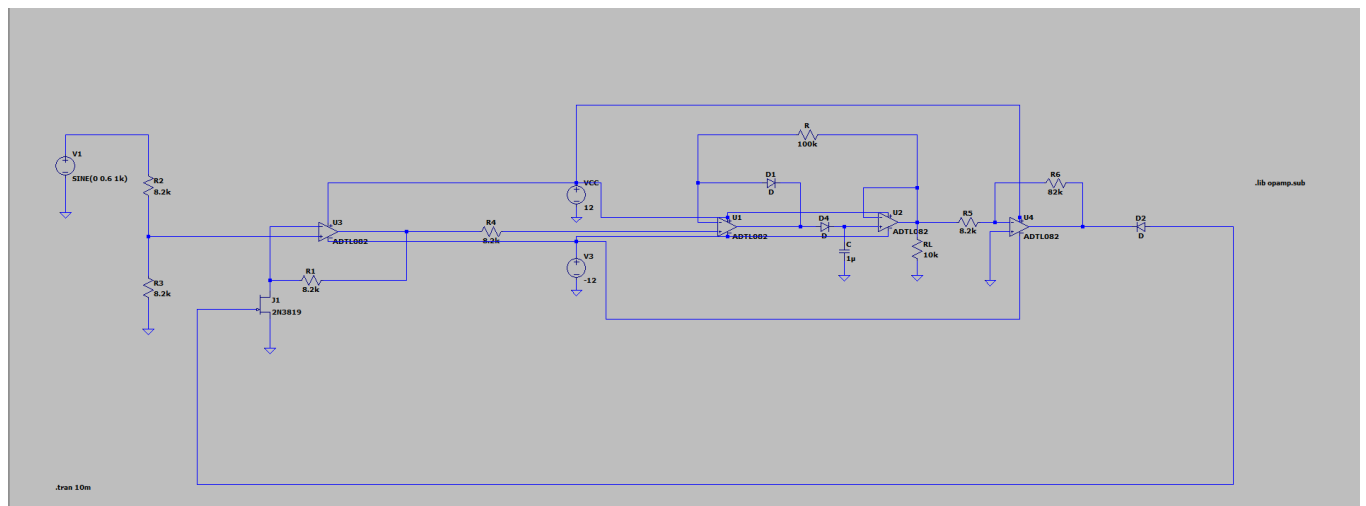
Precision peak detector:



Variable Gain Amplifier:



AGC SYSTEM:



Simulation Procedure

1. Designed the JFET-based VGA.
2. Verified gain variation for different gate-control voltages.
3. Implemented the precision peak detector.
4. Added an inverting control stage.

5. Connected the feedback path to the JFET gate.
6. Performed transient analysis in LTspice.
7. Varied the input amplitude from 80 mV to 600 mV while keeping the frequency constant at 1 kHz.
8. Observed the input signal, output signal, gate voltage, and regulated output amplitude.

Results and Observations

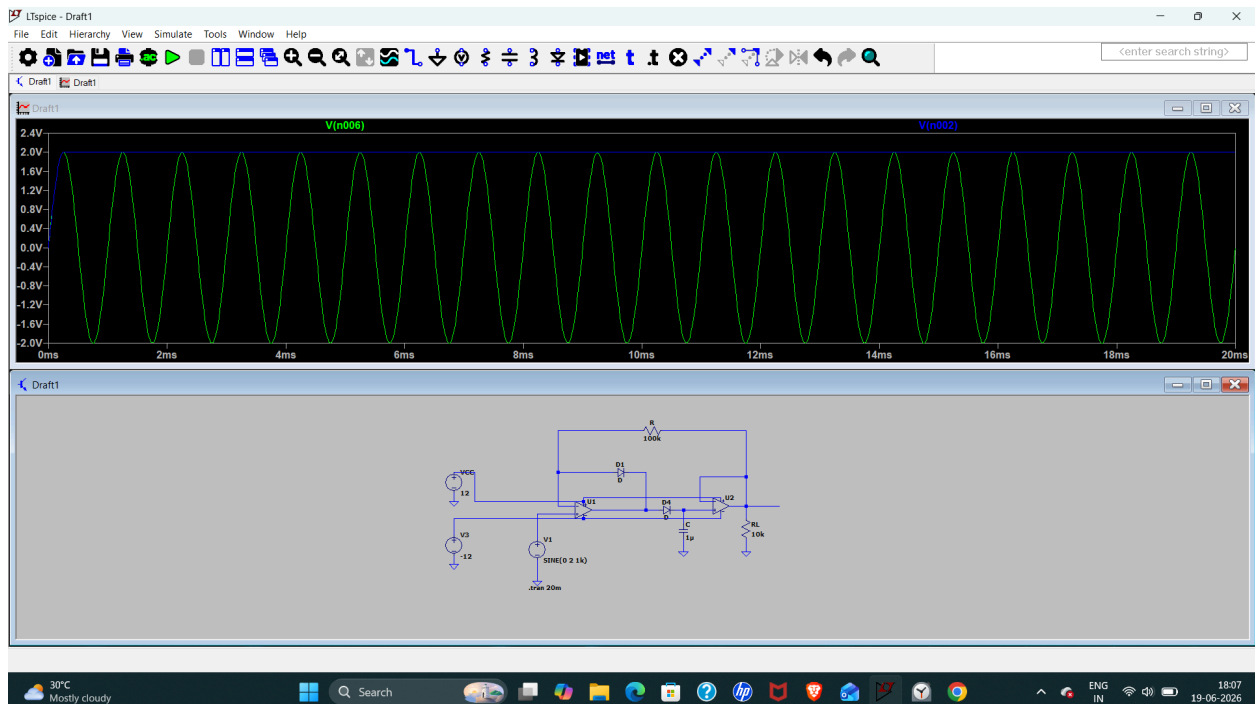
The Variable Gain Amplifier exhibited controllable gain by varying the JFET gate voltage from 0 V to approximately -3 V.

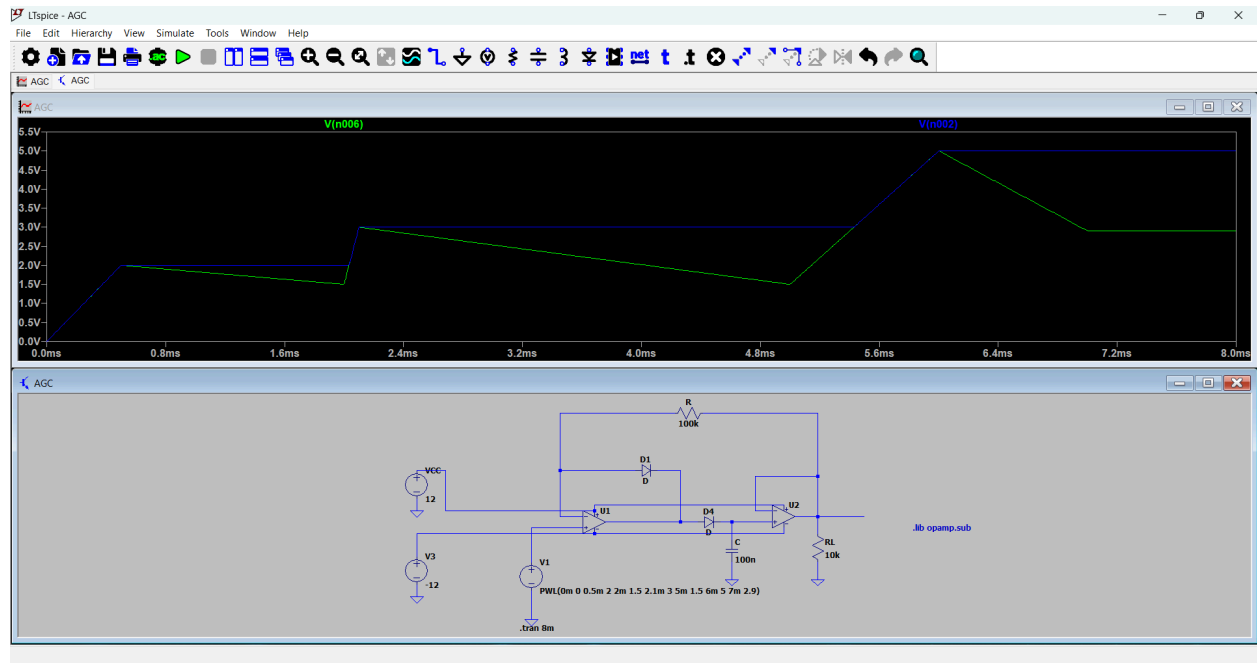
The AGC successfully adjusted the amplifier gain according to input amplitude variations.

For an input amplitude ranging from 80 mV to 600 mV ($7.5\times$ variation), the output remained regulated within approximately 140 mV to 300 mV.

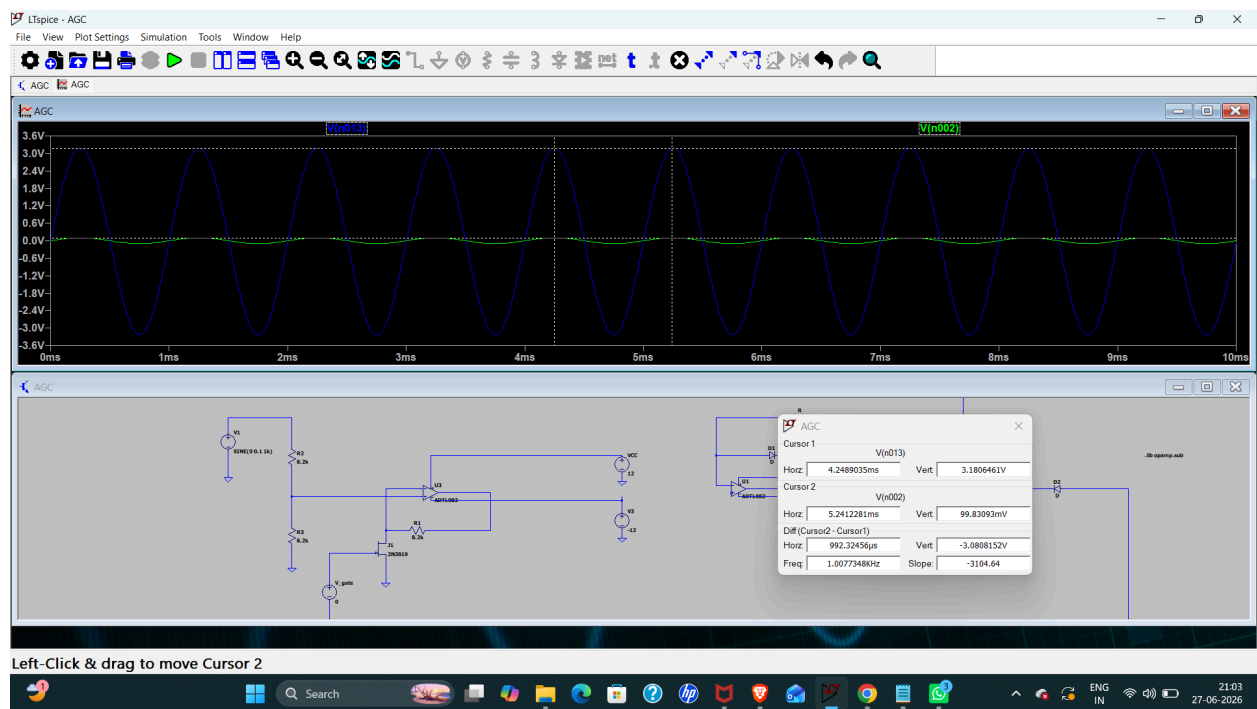
The peak detector generated a stable envelope voltage, while the control amplifier produced the required negative gate voltage for automatic gain adjustment.

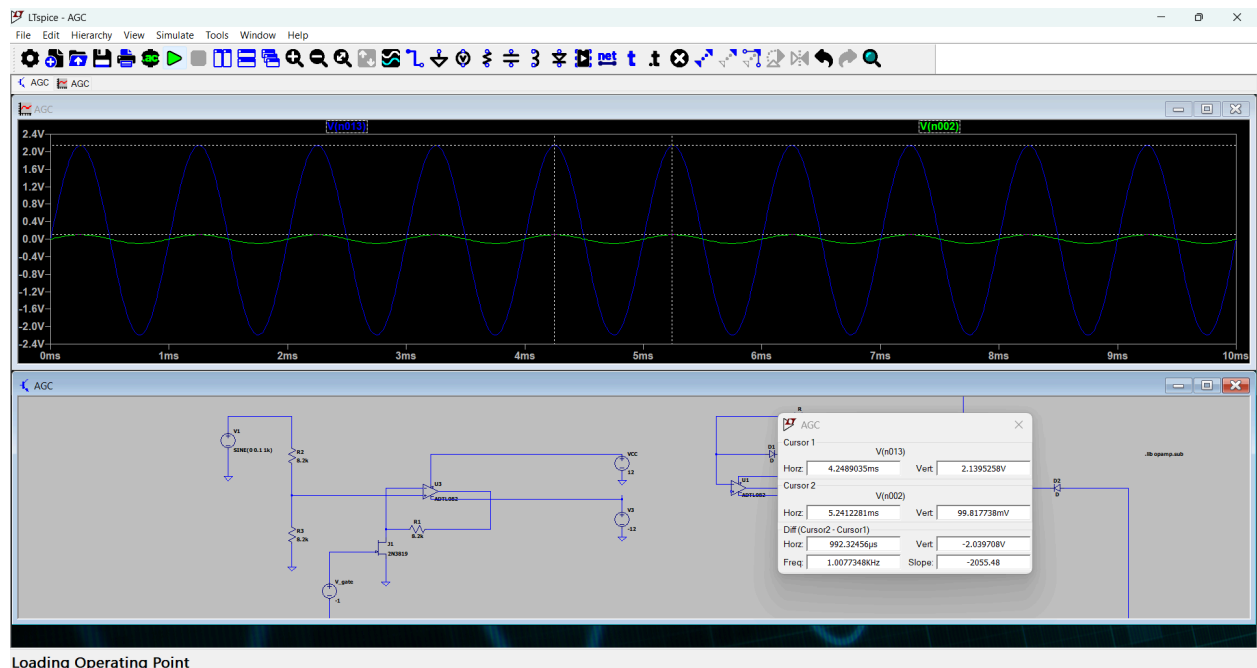
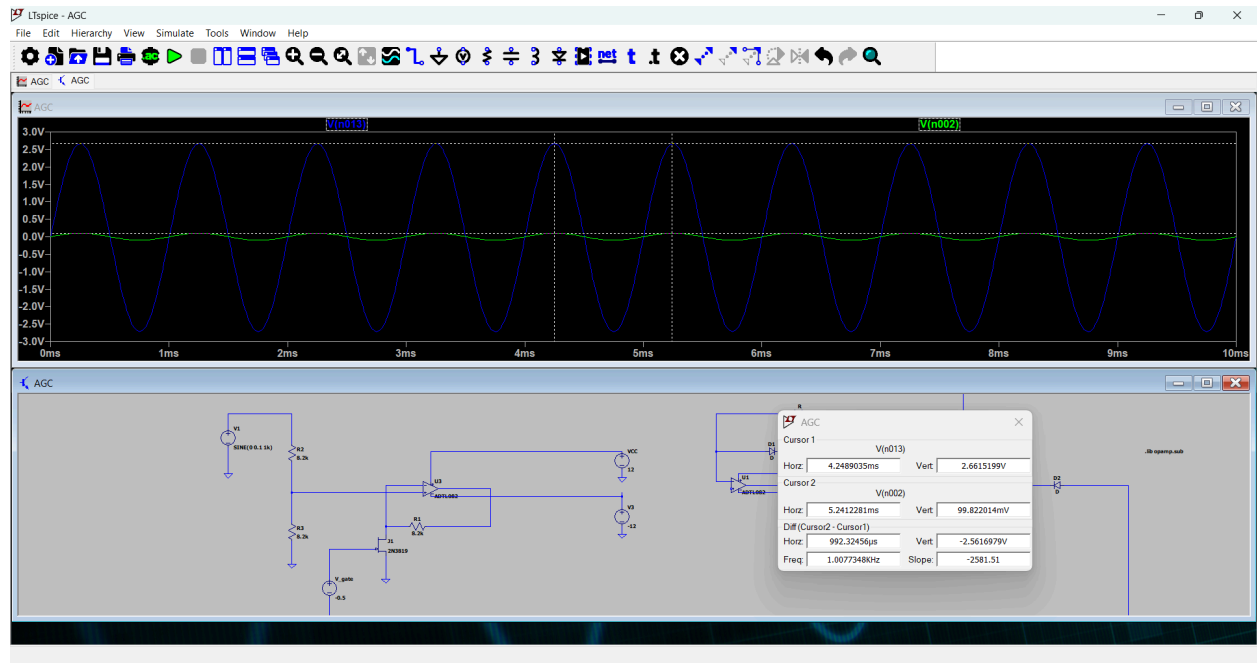
Peak detector simulation results:



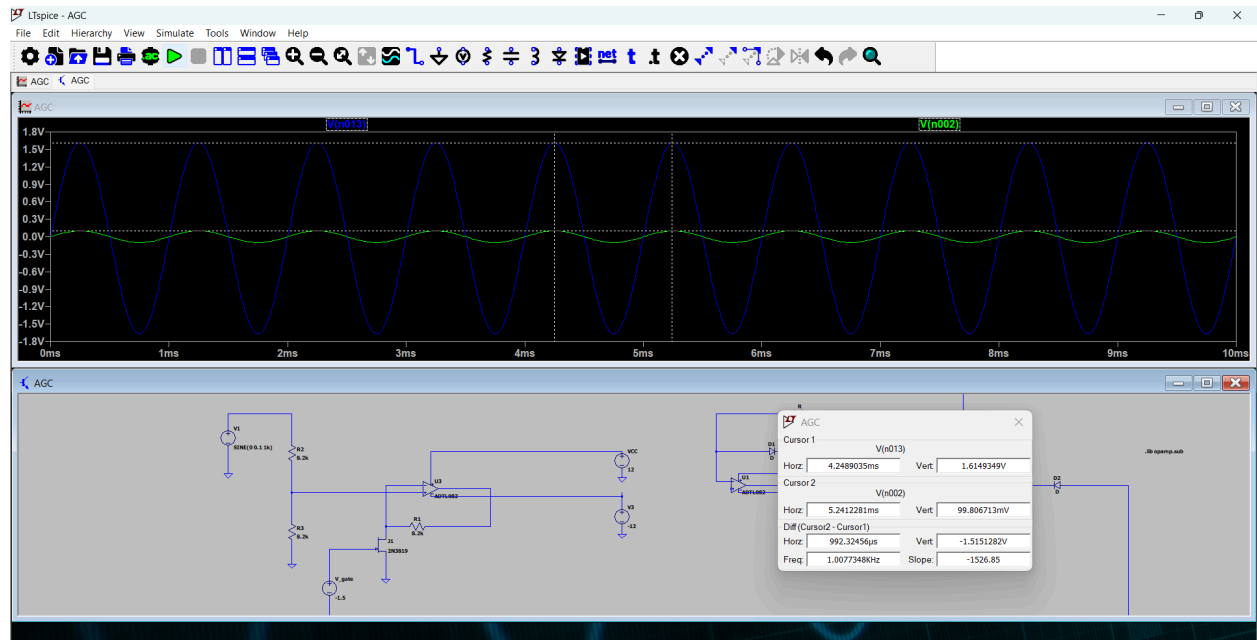


Variable Gain Amplifier Simulation Results:

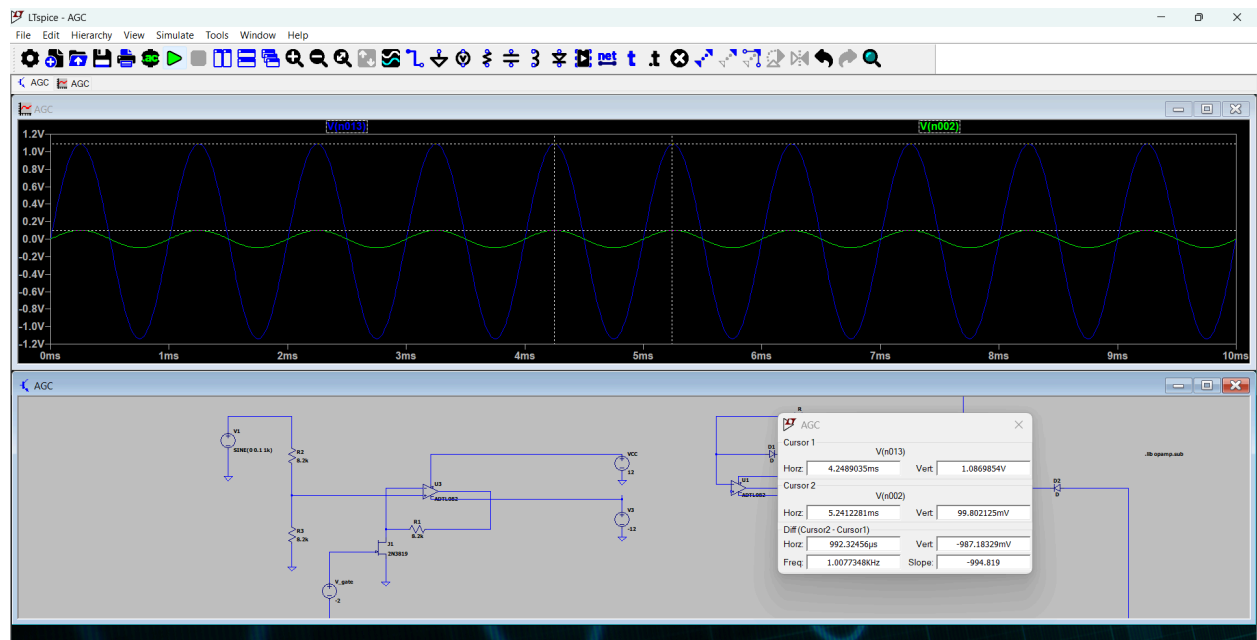




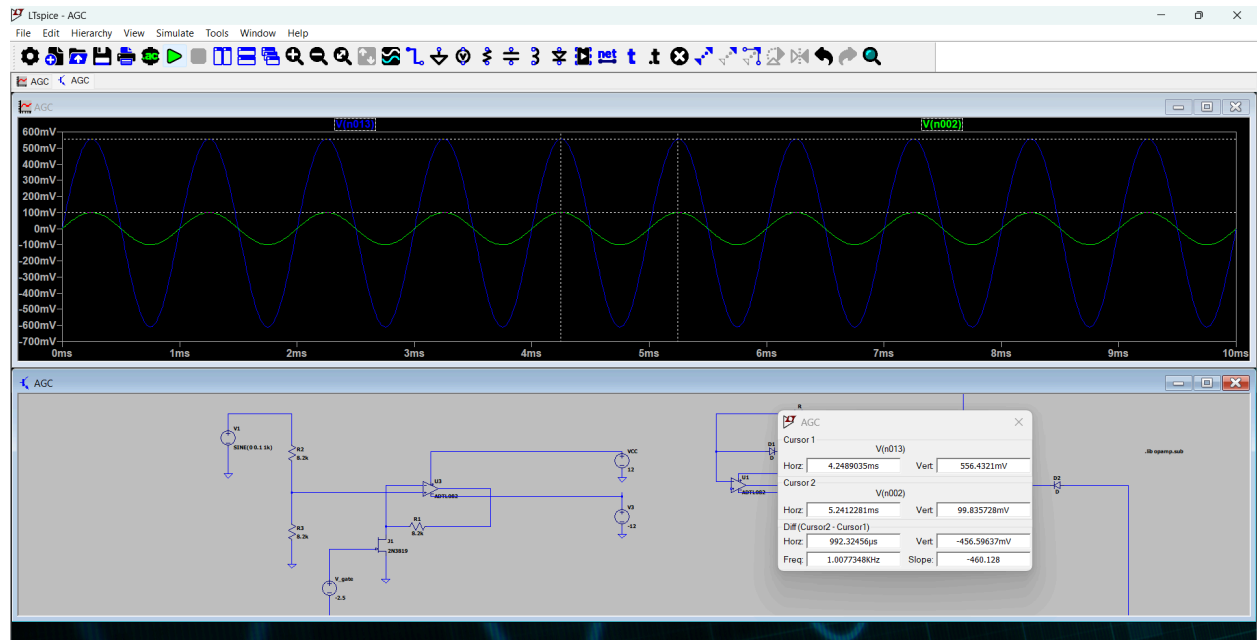
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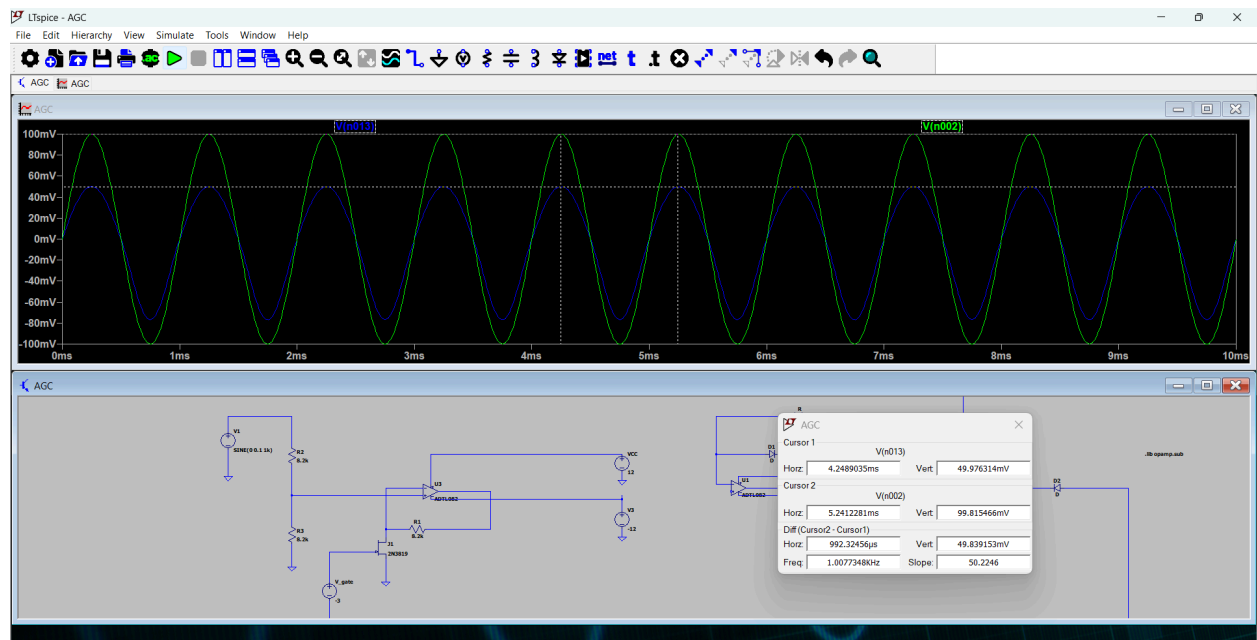
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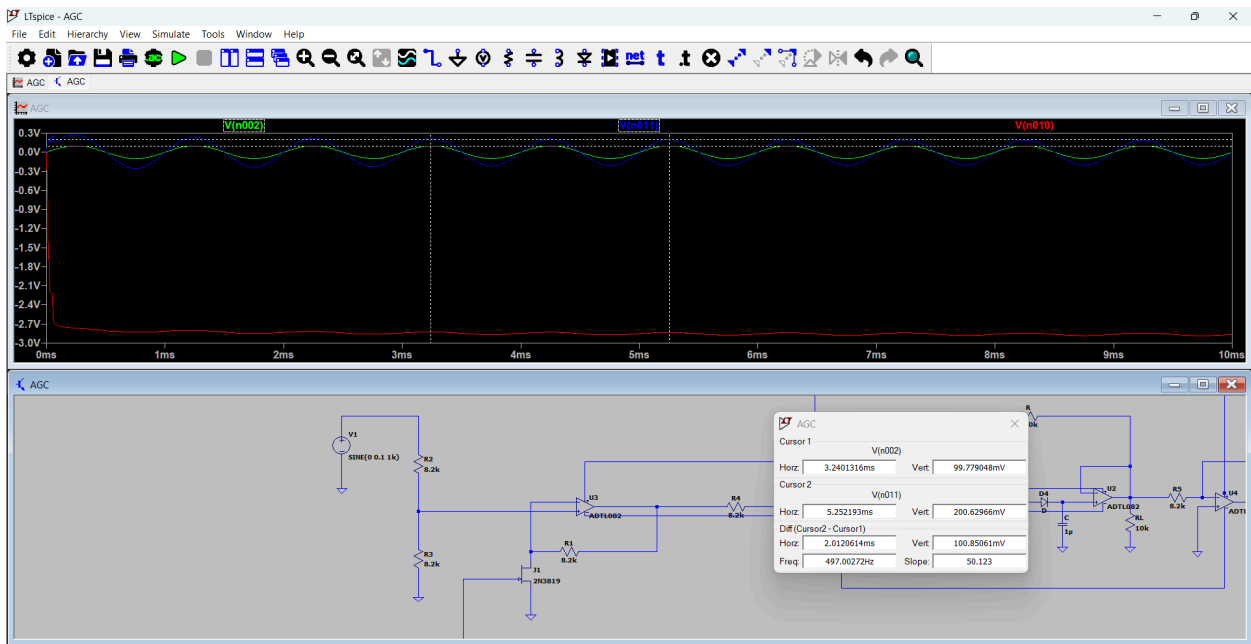


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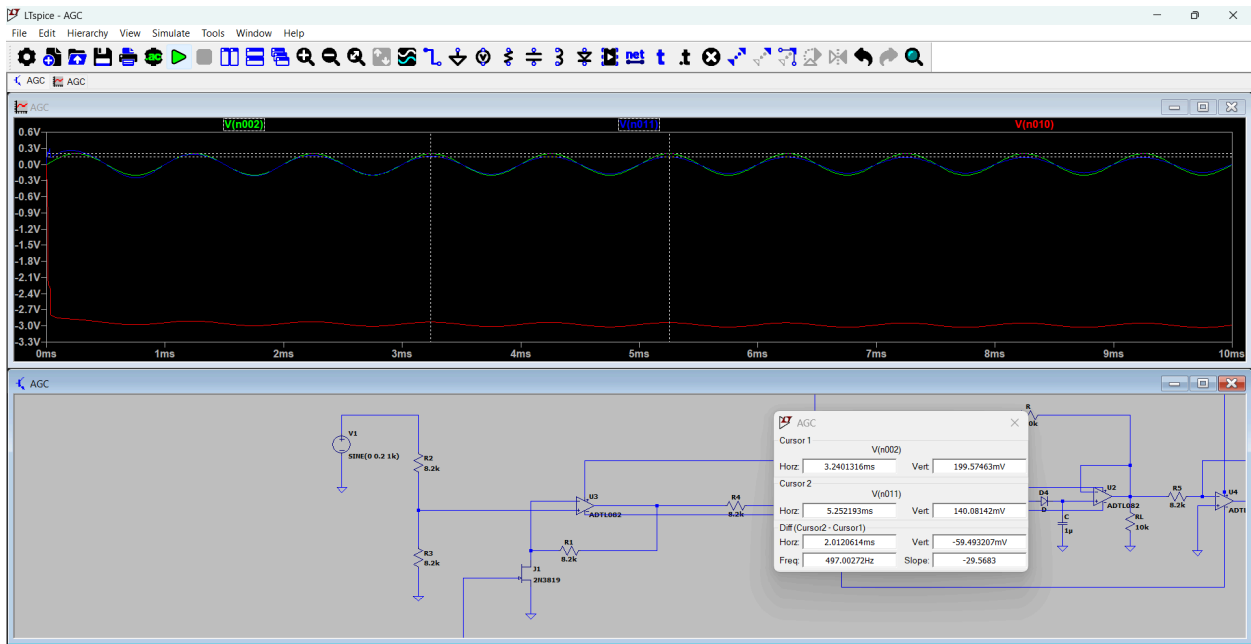


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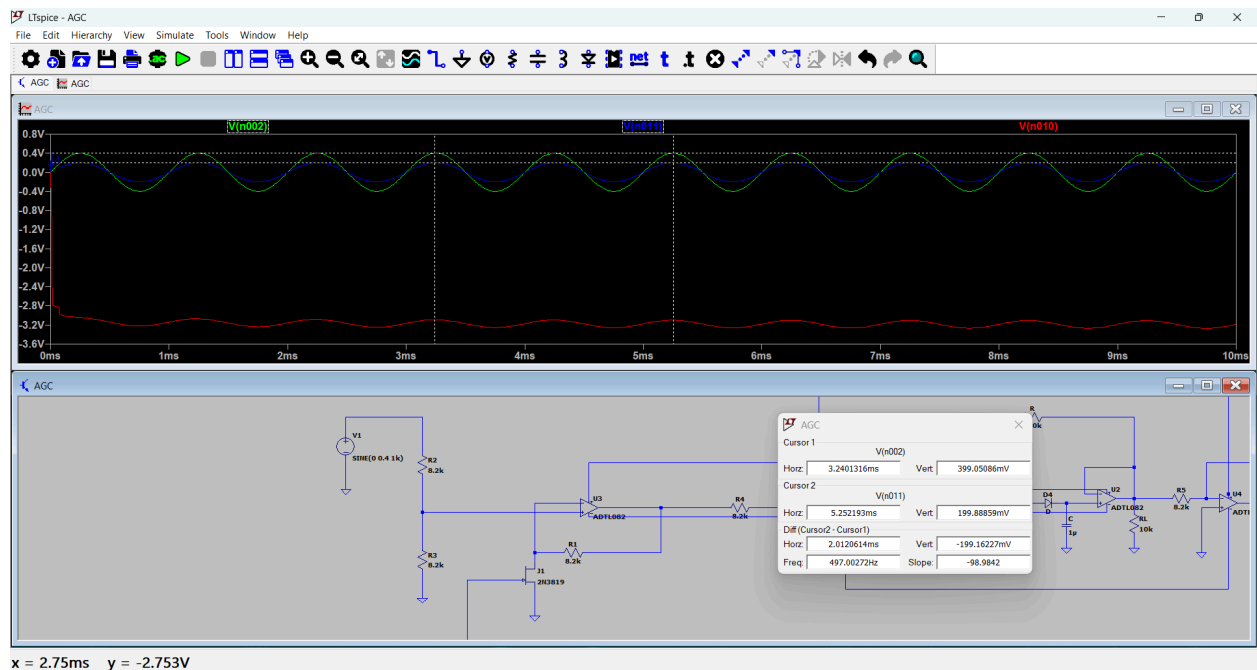
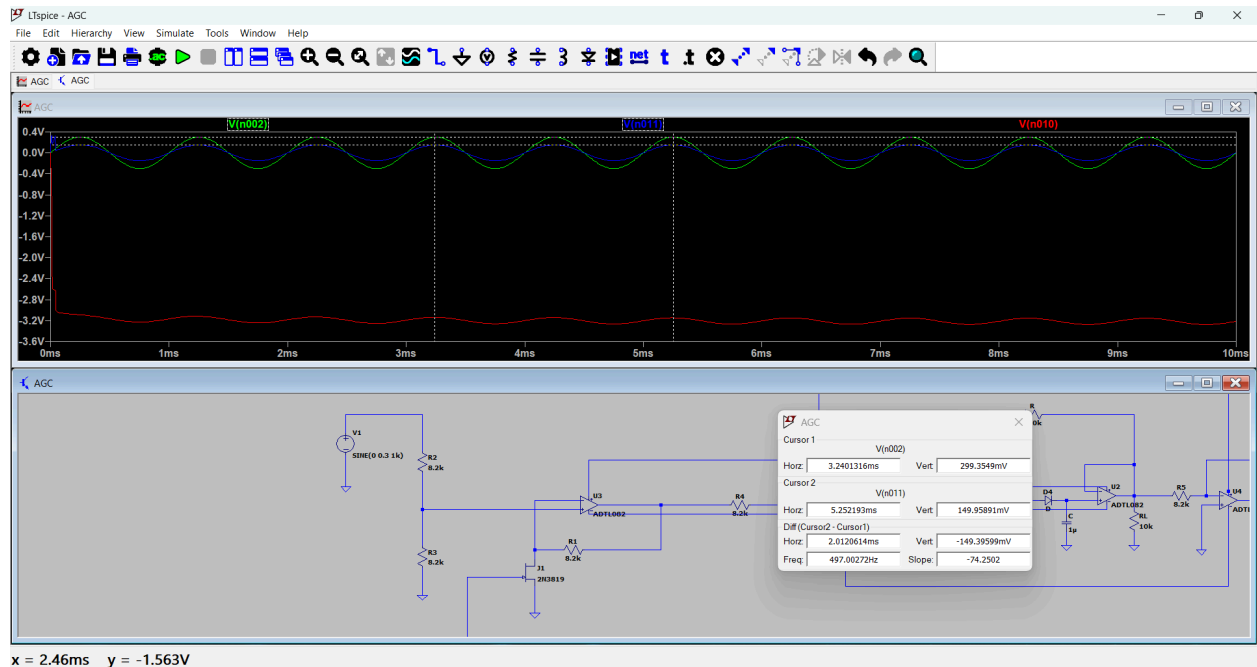
AGC simulation results:

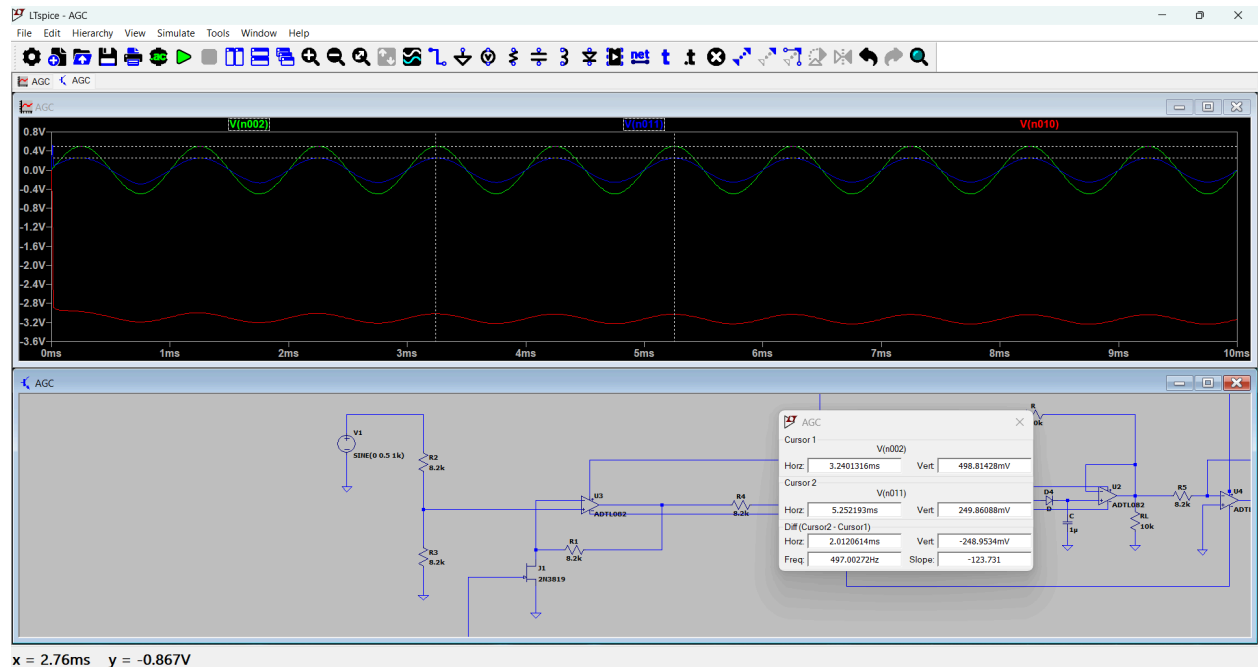


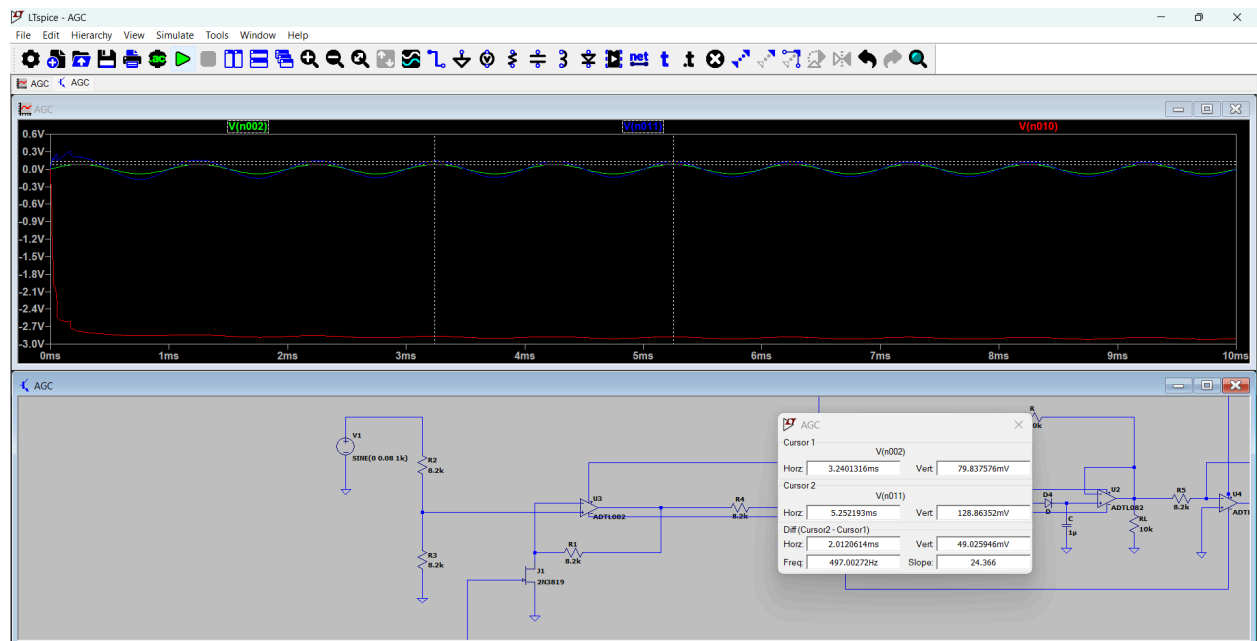
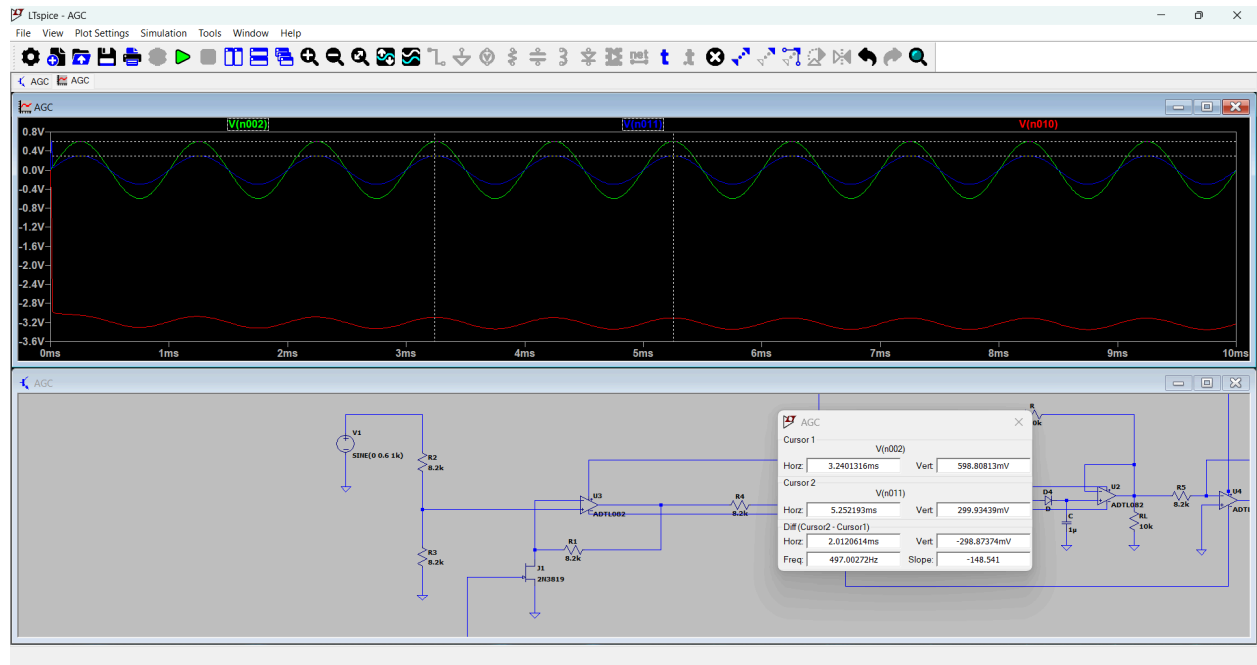
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Key Takeaways

- Learned the operation of JFETs as voltage-controlled resistors.
- Understood closed-loop negative feedback in AGC systems.
- Studied precision peak detection and envelope extraction techniques.
- Gained practical experience in analog circuit simulation using LTspice.
- Analyzed the relationship between JFET gate voltage, channel resistance, and amplifier gain.

Conclusion

A closed-loop Automatic Gain Control system was successfully designed and simulated in LTspice using a JFET-based Variable Gain Amplifier, precision peak detector, and feedback control stage.

The system demonstrated automatic gain regulation by maintaining the output amplitude within approximately 140 mV–300 mV for input amplitudes varying from 80 mV to 600 mV at 1 kHz, validating the effectiveness of the AGC feedback loop.

References

- ADTL082 Datasheet
- 2N3819 JFET Datasheet
- LTspice XVII User Guide
- Sedra & Smith – Microelectronic Circuits
- Analog Devices and Texas Instruments application notes on Automatic Gain Control